

Trustworthy and Explainable Agents for Phishing Detection with Grounded Cross-Model Evidence Agreement

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Abstract

Large language models (LLMs) now flag phishing websites with high accuracy, but a deployed detector consumes a *label* and often discards the *reason*. For actions such as browser warnings, blocklist entries, and take-down requests, operators need a reason they can check and a criterion for when to trust the verdict automatically. Label confidence and majority voting miss a common failure: several models can agree on the right verdict while relying on different, or even fabricated, evidence. We propose PhishProof, a multi-agent selective detector that reads reliability from grounded evidence agreement rather than from labels. A small panel of diverse LLM evidence agents investigates a page, plans which URL, DOM, certificate, redirect, screenshot, and logo signals to inspect, grounds each claim with verification tools, and records the results as a shared set of typed evidence cues. PhishProof acts only when the agents agree on cues that the page itself confirms; otherwise it abstains, returning the agreed, verified cues as the explanation when it does act. A small held-out labeled split calibrates the score to a probability of correctness and fixes the operating point. Because its agents invoke existing reference-based and LLM detectors as verification tools, PhishProof is competitive with those specialized detectors on full-coverage accuracy while adding the calibrated abstention and verifiable explanation they lack. On a phishing benchmark assembled from public feeds and the Tranco top sites, PhishProof lowers AURC by 35% relative to the strongest label-agreement baseline, raises coverage at 99% selective accuracy from 0.48 to 0.71, and abstains on 78–83% of the right-label, wrong-reason and ungrounded regions that baselines act on confidently.

Keywords: phishing detection, LLM agents, tool-augmented agents, evidence cues, selective prediction, cross-model evidence agreement, explanation faithfulness, abstention, trustworthy ML

1. Introduction

A phishing website is a consequential object: when a detector marks a page as phishing, the verdict drives a blocklist entry, a browser warning, or a take-down request that removes a live service from the web. Large language models have raised the accuracy of such detectors sharply, reading a page’s URL, DOM, rendered screenshot, and certificate to decide whether it impersonates a brand it does not own [1, 2]. Yet accuracy is only half of what a deployment needs. The analyst who acts on a verdict, the operator who must justify a take-down, and the user who sees a warning all need a *reason* they can check – and they need to know *when* the detector is trustworthy enough to act on automatically. For a decision that can be contested, the bottleneck is no longer whether the model is usually right, but whether a particular verdict can be *trusted and explained*.

Existing detectors approach this only indirectly. One line runs a single LLM over the page and reports a class probability or a verbalized confidence [1]; reliability is then a property of that one model’s calibration [3]. A second line aggregates several models or several samples – self-consistency over one model [4], majority voting over an ensemble [5], or weighted label agreement in the style of crowd-sourcing and label-free

evaluation [6, 7] – and reads reliability off how often the votes concur. A third, recent line builds multi-agent detectors whose agents specialize on different signals and whose rationales are merged into a single explanation [8, 9]. Together these families cover much of the space of accuracy and aggregation.

They share a blind spot, visible in a routine case (Fig. 1). A page impersonating a bank is shown to a panel of three diverse models; all three return “phishing,” so every aggregation method above reports high confidence. But the models reached that verdict for different reasons: one cited the morphed logo, one cited a credential form posting off-domain, and one cited a certificate mismatch that does not actually hold on the page. First, the detectors emit a label and discard these reasons, so nothing auditable reaches the operator (L1). Second, reliability is read at the *label*, where the three agree, and is therefore blind to the fact that they *disagree on the evidence* – the verdict is right for the wrong reason (L2). Third, where the models do agree on a reason, that agreement can be a shared hallucination of evidence that is not on the page (L3). A page that is right for the wrong reason is not a curiosity: it is exactly the page a small change to the cited cue would flip, and exactly the page an operator should not act on blindly.

Two properties of phishing, not shared by generic question answering, make a direct remedy possible. The evidence a detector relies on is *objectively checkable*: a form-action domain,

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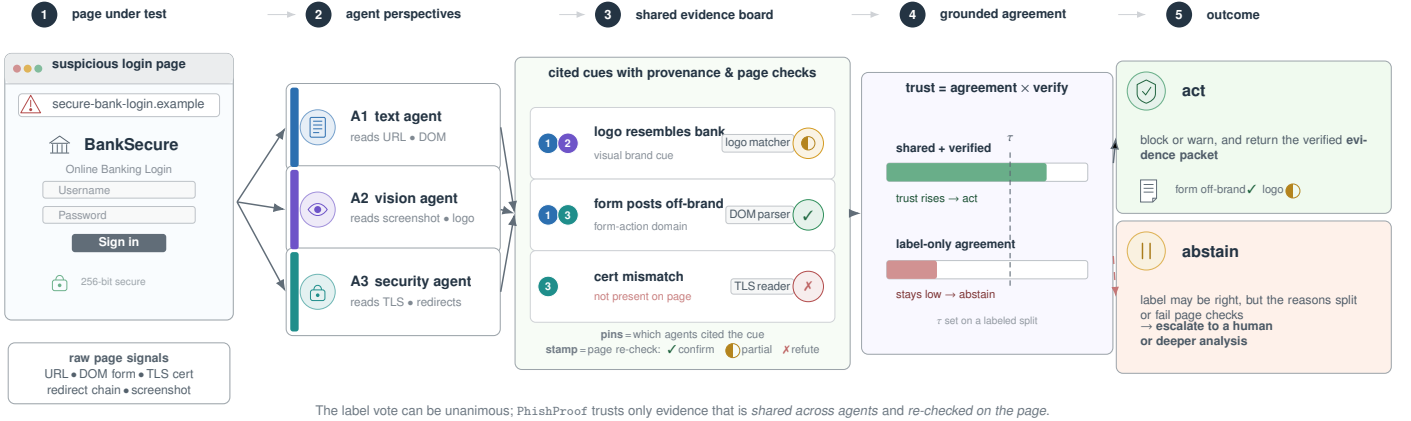


Figure 1: PhishProof as a qualitative evidence story. Three agents inspect the same page from complementary views, then place their cited cues on a shared evidence board. Colored pins show provenance, while tool badges mark cues that are confirmed or rejected by the page itself. PhishProof acts when the shared reasons survive these checks and returns them as an evidence packet; when agents agree on the label but their reasons split or fail verification, it abstains.

a certificate organization, a redirect target, and a rendered logo can each be re-derived from the page itself. And labeled pages are cheap to assemble: public feeds publish freshly confirmed phishing URLs and the Tranco list [10] supplies benign sites, so a detector’s operating point can be set on a held-out labeled split rather than guessed. Crucially, the evidence that decides a phishing verdict is *heterogeneous*: it spans several model types (text LLMs and a vision-language model), many signals (the URL, DOM, certificate, redirect chain, logo, and credential intent), two modalities, and the verification mechanisms that confirm each one. Deciding which of these to gather for a given page and how to combine them is a *planning-and-aggregation* problem that suits an agent rather than a fixed pipeline – and modern LLMs can act as such agents, emitting typed outputs and calling tools to gather and verify their own evidence. The opportunity is to stop treating reliability as a property of labels and start treating it as a property of *evidence that several independent agents plan, gather, verify, and agree on, and that the page confirms*.

To this end, we present PhishProof, a multi-agent phishing detector that trusts a verdict in proportion to grounded cross-model agreement on the evidence. A small, deliberately diverse panel of LLM *evidence agents* investigates the page; rather than emit a bare label, each agent *plans* which signals and modalities to examine, gathers the evidence and grounds it by calling verification *tools* – a DOM parser, a certificate reader, a redirect tracer, a logo matcher, and an existing reference-based or LLM brand detector – then records what it found in a shared, per-page set of typed *evidence cues*, each carrying the asserting agent and the confirming tool. PhishProof *aggregates* the agents’ grounded findings, measures agreement at the level of the cited *evidence* rather than the label, weights it by whether each agreed cue verifies against the page, and combines the two into a single Grounded Evidence-Agreement score. It acts only when that score is high – returning the agreed, verified cues as a checkable explanation – and abstains otherwise, escalating the page to deeper analysis or a human. A calibrator fit on a held-out labeled split maps the score to a probability of correctness

so the operating point is set against real outcomes rather than guessed. The novelty over prior multi-agent detectors is the unit of agreement: not the label, and not one model’s introspective faithfulness, but a set of evidence cues that is simultaneously cross-agent-agreed and tool-verified.

Realizing this raises three difficulties that the design must rule out. Agreement over structured evidence must be defined so that it is not label agreement in disguise – two models can name the same brand while resting on different cues. Shared hallucination must be caught cheaply, without trusting the models to police themselves. And the agent panel must stay small, since every added agent is a round of model calls and tool calls, so the gain has to come from *diversity* rather than count.

We make the following contributions.

- **Failure mode.** We identify the *right-label, wrong-reason* failure of multi-model phishing detectors and cast trustworthy detection as selective prediction over an evidence-agreement signal (§3).
- **Method.** We introduce PhishProof, a panel of tool-using evidence agents that cite a shared set of typed, verified evidence cues, and its Grounded Evidence-Agreement score, which measures cross-agent agreement over those cues and turns it, calibrated on a held-out labeled split, into a selective act/abstain decision with the agreed, verified cues as the explanation (§4).
- **Evidence.** On a phishing benchmark assembled from public feeds and the Tranco top sites, PhishProof is competitive with specialized reference-based and LLM detectors on full-coverage accuracy – it invokes them as verification tools – and improves the risk-coverage trade-off over single-model confidence, self-consistency, and label voting: AURC falls by 35% relative to the strongest baseline and coverage at 99% selective accuracy rises from 0.48 to 0.71 (§5).
- **Analysis.** We isolate the contribution of evidence-level agreement, tool-grounding, agent diversity, and calibration. The right-label, wrong-reason and ungrounded regions flip under evidence-targeted perturbations 41.6%

and 47.9% of the time, and PhishProof abstains on 78–83% of them (§5).

The rest of the paper is organized as follows. §2 positions PhishProof against label-level reliability and multi-agent detection. §3 formalizes grounded evidence-agreement and the selective objective. §4 details the evidence agents, the cues they cite, the score, and the calibrated decision rule, and §5 reports the evaluation.

2. Related Work

LLM and reference-based phishing detection. The dominant modern detectors decide whether a page impersonates a brand it does not own. Reference-based systems match a rendered page against a brand library of logos and layouts [11, 12, 13], and recent work refreshes that reference set online [14] or replaces it with an LLM that recognizes the brand and checks the domain directly [1, 2]. A parallel line treats the page as URL or image features for a learned classifier [15], surveyed broadly [16]. These systems optimize and report *aggregate* accuracy. PhishProof builds on this line rather than competing with it: its agents invoke these detectors as verification tools, so it is *competitive* on full-coverage accuracy (Tab. 4) while adding what a bare detector lacks – a *per-verdict* decision about whether the verdict can be trusted, acted on automatically, and explained (§4.4).

Multi-agent and ensemble phishing detection. Closest to us are detectors that run several models. Majority voting over LLMs has been studied directly for phishing [5]; multi-agent systems assign agents to different signals and either debate to a verdict or merge their rationales into one explanation [8, 9], sometimes backed by an episodic memory of past cases [17]. Unlike these systems, which aggregate or consolidate at the level of the *label* or a fused rationale, PhishProof measures agreement at the level of individually cited evidence units ((2)) and treats *disagreement on evidence* as the signal to abstain rather than a conflict to resolve. Its agents are tool-using investigators that each *plan* their own investigation across signals and modalities, and whose tool-grounded findings are *aggregated* into a shared set of cited evidence cues that serves as both the trust signal and the explanation, rather than a fused free-text rationale.

Label-level reliability and label-free evaluation. A verdict’s reliability is usually read from the label distribution: post-hoc confidence calibration [3], agreement across samples of one model [4], or agreement across many models, as in crowd-sourcing aggregation [6], agreement-on-the-line under shift [18], and trust calibration for phishing detectors [7]. Where these estimate reliability from *how often labels concur*, PhishProof estimates it from *whether the reasons concur and hold*, which separates pages on which the panel agrees by coincidence from those on which it agrees because the page truly carries the evidence ((4)).

Explanation faithfulness. Causal audits show that an LLM’s stated rationale is often not the driver of its decision [19], and multi-agent debate has been used to *improve* the faithfulness

Table 1: Capability comparison for per-verdict trust in multi-model phishing detection. ●: yes; ○: partial; ○: no. Columns are defined in the body; full-coverage detection accuracy is reported separately in Tab. 4.

Approach	Evid.- level	Page- grounded	Calib.- trust	Selective abstain	Explan. output
Phishpedia [12]	○	●	○	○	●
PhishLLM [1]	○	○	○	○	○
Single conf. [3]	○	○	○	●	○
Self-consist. [4]	○	○	○	○	○
Majority vote [5]	○	○	○	○	○
Label agree. [6]	○	○	○	○	○
MultiPhishGuard [9]	○	○	○	○	○
Faithfulness audit [19]	○	○	○	○	○
PhishProof (ours)	●	●	●	●	●

of generated explanations [20]. Such audits operate on a single model and require counterfactual re-execution. In contrast, PhishProof uses cross-model agreement on grounded evidence as an inexpensive faithfulness *proxy*: a reason that several independent models cite and that the page confirms is, by construction, less likely to be a post-hoc rationalization, at no extra re-execution cost.

Selective prediction. We build on selective classification, which trades coverage for accuracy via an abstention rule [21], and on distribution-free uncertainty quantification [22]. Building on this foundation, PhishProof supplies the missing ingredient for the phishing setting: an abstention score derived from grounded cross-model agreement, calibrated on a held-out labeled split so the risk–coverage operating point is set against real outcomes.

Position and scope. PhishProof contributes a single new object, the Grounded Evidence-Agreement score, and the selective detector built on it. Tab. 1 positions it against one representative per family along five capabilities – evidence-level agreement, page-grounding, calibrated trust, selective abstention, and a verifiable explanation as output – and PhishProof is the only entry that supports all five. We focus on per-verdict trust and explanation for a multi-model phishing detector; the framework is agnostic to the specific base models and evidence schema. We do *not* claim (i) a new base detector or a leaderboard win on full-coverage accuracy – PhishProof reuses existing detectors as tools and is *competitive* with them (Tab. 4), contributing the per-verdict trust and explanation layer on top; (ii) robustness to an adversary who controls the panel agents or corrupts the tools’ inputs, a harness-level threat we treat as out of scope; or (iii) coverage guarantees on fully cloaked pages that carry no verifiable cues, where the detector necessarily abstains rather than errs.

3. Problem Formulation

Setting. A detector receives a webpage x , represented by its URL, DOM, rendered screenshot, TLS certificate, and redirect chain, and must decide whether x is phishing. We assume access to a panel $\mathcal{M} = \{m_1, \dots, m_M\}$ of M diverse LLM evidence agents (different model families, including at least one vision-language model that reads the screenshot), each able to call a set of verification *tools*, and to a fixed, typed evidence schema $\mathcal{U}$.

Table 2: Summary of important notations.

Symbol	Meaning
x	a webpage under test (URL, DOM, screenshot, cert, redirect)
$\mathcal{M} = \{m_1, \dots, m_M\}$	panel of M diverse LLM evidence agents ($M=3$)
$\mathcal{U}$	fixed typed evidence-cue schema (§4.1)
$E_m \subseteq \mathcal{U}$	grounded cues agent m cites on x
C	<i>consensus</i> cue set (cues a majority of agents assert)
$g(u, x) \in [0, 1]$	tool that re-derives cue u 's type from x
$A(x) \in [0, 1]$	evidence-level agreement over the shared cue set
$G(x) \in [0, 1]$	groundedness of the consensus cues
$\text{GEA}(x) = A \cdot G$	Grounded Evidence-Agreement score
τ	abstention threshold (acts iff $\text{GEA} \geq \tau$)
$\hat{\rho} = \kappa(\text{GEA})$	calibrated map $\text{GEA} \rightarrow \mathbb{P}(\text{correct})$, fit on a labeled split

Table 3: The traceable thread: each limitation (§1) becomes a requirement and a named component.

Limitation	Requirement	Component
L1 label-only, unverifiable output	R1 emit structured, checkable evidence	C_1 Agentic evidence elicitation (§4.1)
L2 reliability read at the label, blind to wrong reasons	R2 measure agreement at the <i>evidence</i> level	C_2 Evidence agreement A (§4.2)
L3 shared hallucination of false evidence	R3 ground cited evidence to the page	C_3 Tool-grounding G (§4.3)

and a calibrator $\kappa : \text{GEA} \mapsto \hat{\mathbb{P}}(\hat{y} = y)$ fit on a held-out labeled split so the operating point τ realizes a target risk. *Outputs* per page: a verdict or abstention, the calibrated trust $\hat{\rho} = \kappa(\text{GEA})$, and, when acting, the consensus cue set $C(x)$ as the explanation.

Framework overview. Fig. 2 maps the requirements to the components evaluated in §5: agentic evidence elicitation – agents call tools to cite cues (C_1 , **R1**), evidence agreement over the shared cue set (C_2 , **R2**), and tool-grounding (C_3 , **R3**); a calibrated selective rule (C_4) then turns the score into an act/abstain decision. §4 details each; Tab. 3 is the thread we follow through to the ablation.

4. The PhishProof Detector

PhishProof turns the score of (4) into a calibrated, selective verdict through the components of Fig. 2. We describe each, then give the end-to-end procedure and its cost.

4.1. Evidence agents and the shared cue set

Each member of the panel is an *evidence agent*: prompted with the page, it *plans* which signals and modalities to examine, calls the verification tools to obtain them, and returns a verdict together with the grounded cues it found, drawn from the fixed schema $\mathcal{U}$ and emitted by constrained decoding as well-typed (t, v) pairs rather than free text (**R1**). The schema is small and auditable – six types covering the brand claim, the form-action domain, the certificate organization, the redirect target, credential intent, and the logo brand – and is shared by every agent, so their cues are directly comparable. The agents’

A cue $u = (t, v) \in \mathcal{U}$ pairs a type t (e.g. form-action-domain, cert-org, redirect-target, logo-brand, credential-intent) with a value v ; each type has a tool g that re-derives it from x . The agents’ findings are collected into a per-page *shared set of cited cues*, each tagged with the agent that asserted it and the tool that can check it. Notation is summarized in Tab. 2.

Agent outputs and the shared cue set. Run on x , each agent investigates the page and returns a verdict together with the cues it grounds through tool calls:

$$m_i(x) = (y_i(x), E_i(x)), \quad y_i \in \{\text{phish}, \text{benign}\}, \quad E_i \subseteq \mathcal{U}. \quad (1)$$

The agents’ cues form the page’s shared cue set $\bigcup_i E_i(x)$, and the verdict is the majority vote $\hat{y}(x) = \text{maj}_i y_i(x)$. A *consensus* cue is one a strict majority of the agents assert with a consistent value, $C(x) = \{u : |\{i : u \in E_i\}| > M/2\}$; these consensus cues are the evidence PhishProof trusts and returns as the explanation.

Prior formulation and its limits. Existing detectors estimate a per-verdict reliability $r(x)$ from the *label* channel alone – a single model’s confidence, or the concordance of $\{y_i\}$ [3, 5, 6]. This label-centric formulation has three limitations, which we restate as the requirements PhishProof must meet (Tab. 3). It returns no checkable artifact, only $\hat{y}$, so an action cannot be justified (**L1**; **R1**: emit structured evidence). It is a function of $\{y_i\}$ only, so it cannot distinguish a panel that agrees *because* the page carries the evidence from one that agrees on the label while resting on different cues (**L2**; **R2**: score agreement on evidence). And it treats a cited reason as true, so a reason all models hallucinate is rewarded (**L3**; **R3**: ground evidence to the page).

Grounded evidence-agreement. We replace label reliability with a score on the *evidence* channel. Define the evidence-level agreement and the groundedness of the consensus,

$$A(x) = \frac{|C(x)|}{\left| \bigcup_i E_i(x) \right|}, \quad (2)$$

$$G(x) = \frac{1}{|C(x)|} \sum_{u \in C(x)} g(u, x), \quad (3)$$

where $g(u, x) \in [0, 1]$ is the tool that re-derives cue u ’s type from x (deterministic for form-action, cert-org, redirect; perceptual for logo-brand); $G(x) = 0$ when $C(x)$ is empty. The *Grounded Evidence-Agreement* score is their product,

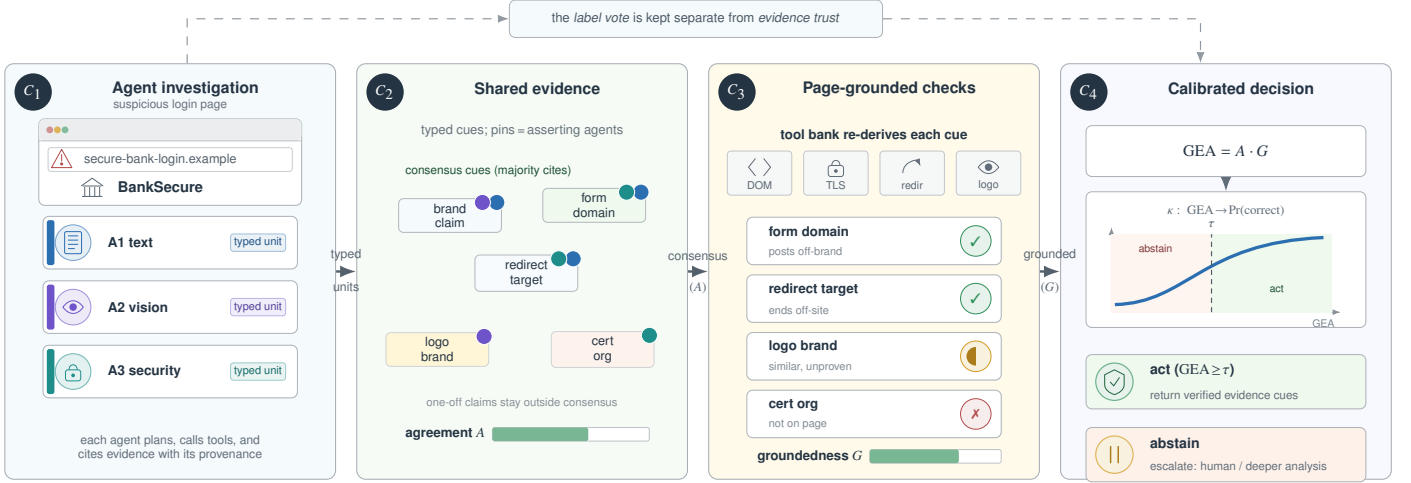
$$\text{GEA}(x) = A(x) \cdot G(x) \in [0, 1]. \quad (4)$$

Agreement on cues the page does not confirm ($G \rightarrow 0$) and label agreement without evidence consensus ($A \rightarrow 0$) both drive GEA down, directly discharging **R2** and **R3**.

Objective. PhishProof is a selective detector: it acts with $\hat{y}(x)$ when $\text{GEA}(x) \geq \tau$ and abstains ($\perp$) otherwise. For a threshold τ , *coverage* and *selective risk* are

$$\phi(\tau) = \mathbb{P}[\text{GEA} \geq \tau], \quad r(\tau) = \mathbb{P}[\hat{y} \neq y \mid \text{GEA} \geq \tau], \quad (5)$$

with y the true label. We seek a score whose risk–coverage curve $r(\phi)$ lies as low as possible (small area under it, AUC),



PhishProof trusts a verdict only when agents agree on the same evidence and the page re-confirms it; the agreed, verified cues are the returned explanation.

τ, κ fit on a held-out labeled split

Figure 2: PhishProof qualitative pipeline. Agents inspect the same page from complementary views and cite typed evidence cues into a shared set. The system trusts a verdict only when the agents converge on the same cues and the tool bank can re-check those cues against the live page. The calibrated selector returns the label with the verified cues when the evidence is shared and grounded; otherwise it abstains, including cases where the label vote is plausible but the reasons are thin or unverified. The threshold and calibrator are set on a held-out labeled split.

cues are collected into the page’s *shared cue set*: each cue is typed, tool-checkable, and tagged with the agent that asserted it, so the verdict rests on a list of cited, verifiable reasons – a brand claim paired with off-brand infrastructure – rather than an opaque score. Agent *diversity* is a design choice, not an afterthought: the three agents come from different model families and at least one is a vision-language model that reads the screenshot, so that agreement on the cues reflects independent evidence rather than a shared prior (its value is isolated in §5).

4.2. Evidence agreement

To *aggregate* and compare cues across agents we first *normalize values*: domains are reduced to their registrable eTLD+1, brands to a canonical lexicon entry, so that “paypal-secure.com” and “paypal-secure.com/login” count as the same form-action cue. The consensus set $C(x)$ collects the value-normalized cues a strict majority of the agents assert, and the evidence-level agreement $A(x)$ of (2) is the share of all cues in the set that fall in this consensus (R2). Agreement is thus high only when the agents converge on the *same* evidence, and is unaffected by their concurring on the label alone.

4.3. Tool-grounding

A consensus can still be wrong if the agents share a hallucination, so each consensus cue is checked against the page by the tool for its type $g(u, x)$ (R3). The structural tools verify deterministically – the DOM is parsed for the form-action domain, the certificate for its organization, and the redirect chain for its final target – and return 0 or 1. The perceptual tool (logo-brand) matches the rendered logo to the claimed brand and returns a similarity in $[0, 1]$. The brand claim itself is grounded by an existing specialized detector used as a tool – a reference-based visual matcher [12, 13] or an LLM brand-and-domain checker [1]

Algorithm 1 PhishProof inference on a page x

- 1: **for** $m_i \in \mathcal{M}$: $(y_i, E_i) \leftarrow m_i(x)$ $\triangleright$ C_1 agents call tools, (1)
- 2: $\hat{y} \leftarrow \text{maj}_i y_i$; $E \leftarrow \bigcup_i E_i$ $\triangleright$ shared cue set
- 3: $C \leftarrow \{u : u \text{ asserted by } > M/2 \text{ agents, value-normalized}\}$
- 4: $A \leftarrow |C| / |E|$ $\triangleright$ C_2 , (2)
- 5: **for** $u \in C$: compute $g(u, x)$; $G \leftarrow \text{mean}_{u \in C} g(u, x)$ $\triangleright$ C_3 , (3)
- 6: $\text{GEA} \leftarrow A \cdot G$; $\hat{\rho} \leftarrow \kappa(\text{GEA})$ $\triangleright$ (4)
- 7: **if** $\text{GEA} \geq \tau$ **return** $(\hat{y}, \hat{\rho}, C)$; **else return** $\perp$

– so the panel *subsumes* the signals of the dedicated phishing detectors rather than re-deriving them, and inherits their detection quality by construction (§5). The groundedness $G(x)$ of (3) averages these over the consensus, and the score $\text{GEA} = A \cdot G$ of (4) collapses whenever the agreed evidence does not survive verification.

4.4. Calibrated selection

Finally, PhishProof turns the score into a decision. Because phishing labels are available at collection time – a flagged URL is confirmed by the public feed it is drawn from [23, 24] – we set the operating point on a small held-out labeled split: we fit a monotonic (isotonic) calibrator $\kappa : \text{GEA} \mapsto \hat{\mathbb{P}}(\hat{y} = y)$ on it and pick the smallest threshold τ meeting a target selective risk. The detector then acts with $\hat{y}(x)$ when $\text{GEA}(x) \geq \tau$ and abstains otherwise, escalating the page to deeper analysis or a human. When it acts, it returns the consensus cue set $C(x)$ as a verifiable explanation.

4.5. Procedure, guarantee, and cost

Walk-through. 1 runs the agent panel once, each agent calling its tools to cite its cues (line 1), takes the majority label (line 2),

and then does the work that distinguishes PhishProof from label-level reliability: it forms the consensus set (line 3), scores³⁹⁰ agreement on that evidence rather than on the label (line 4), checks each agreed cue against the page with its tool so a shared hallucination cannot inflate the score (line 5), and combines the two factors into GEA (line 6). The calibrated trust and the selective decision (line 7) set the operating point against the labeled split. The novelty is concentrated in lines 3–6: the quantity that gates the verdict is grounded agreement on the *cues*, and the returned cue set is simultaneously the trust signal and the explanation.³⁹⁵

Guarantee. Tool-grounding is what rules out shared hallucination, and we can state this precisely.

Assumption 1 (Tool soundness). *For each cue type, the tool $g(u, x) = 1$ if u holds on x and 0 otherwise (perceptual types: $g(u, x)$ is the true match score).*⁴⁰⁰

Proposition 1. *Under Assumption 1, any consensus cue that does not hold on x contributes 0 to $G(x)$; hence $\text{GEA}(x)$ is at most the grounded fraction of the consensus, and $\text{GEA}(x) = 0$ whenever the entire consensus is hallucinated.*⁴⁰⁵

Proof sketch. G is the mean of g over C ((3)); by Assumption 1 a non-holding cue adds 0 to that mean, so $G \leq \frac{|\{u \in C : u \text{ holds}\}|}{|C|}$, and $\text{GEA} = A \cdot G \leq G$ inherits the bound. If no consensus cue holds, $G = 0$ and thus $\text{GEA} = 0$. $\square$

The proposition is modest by design: it bounds, but does not eliminate, gauge error, since a tool is sound for what it checks,⁴¹⁵ but a page can carry true-yet misleading evidence; we report tool precision and recall against gold in §5.

Complexity. Per page, PhishProof costs M agents – each a model call plus the tool calls it issues – and the verification of the $|C|$ consensus cues; the structural tools are⁴²⁰ $O(1)$ DOM/certificate/redirect lookups and the single perceptual check is one embedding comparison, so the cost is dominated by the $M=3$ agents and is independent of the schema size beyond the cues actually asserted. Relative to a single-model detector this is a fixed $M \times$ overhead with no training; under⁴²⁵ selective operation only the abstaining fraction $1 - \phi(\tau)$ is escalated to deeper analysis, so the amortized cost is tuned by the same threshold that sets the risk–coverage operating point (§5).

5. Experiments

We evaluate PhishProof as a selective detector: it may³⁸⁰ act only on pages whose evidence is trustworthy enough, and should abstain otherwise. The four questions below each test⁴³⁵ one method claim from §3:

- **RQ1 – Selective detection** (§5.2): does grounded evidence-agreement give a better risk–coverage trade-off than label-level reliability?
- **RQ2 – Right-label, wrong-reason** (§5.3): does⁴⁴⁰ PhishProof flag the pages a panel labels correctly but for the wrong reason, which the baselines act on?

- **RQ3 – Component attribution** (§5.4): how much do evidence-level agreement, groundedness, and diversity each contribute?
- **RQ4 – Calibration** (§5.5): is the trust score calibrated, so its reported probability of correctness can be acted on?

5.1. Setup

Benchmark and labels. We evaluate on PHISHSEL, a benchmark of roughly 4,000 webpages assembled over a two-week window: live phishing URLs drawn from public feeds [23, 24] and benign pages sampled from the Tranco top-sites list [10], each captured with its URL, DOM, rendered screenshot, TLS certificate, and redirect chain. Labels are available at collection – a feed-listed URL is a confirmed phish and a Tranco top site is benign – so the benchmark needs no delayed labeling. We split the pages into a small labeled *calibration* split and a disjoint held-out *test* split, fit all thresholds and the calibrator on the former, and report every number on the latter.

Agents, tools, and baselines. The default panel is $M=3$ deliberately diverse *evidence agents*: a frontier text LLM, a second text LLM from a different family, and a vision-language model that reads the screenshot, each able to call the verification tools of §4.3. All methods receive the same captured page snapshot and no label at inference time. PhishProof emits constrained evidence cues in the schema of §4.1, normalizes values as in §4.2, and verifies only the consensus cues before calibration. We compare against six reliability baselines that share this panel or a member of it:

- *(B1) Single-model confidence* [3]: the strongest single model with temperature-scaled softmax confidence.
- *(B2) Verbalized confidence*: the same model asked to state its confidence.
- *(B3) Self-consistency* [4]: one model, $k=5$ samples, agreement as the score.
- *(B4) Majority vote* [5]: panel label agreement.
- *(B5) Label-agreement* [6]: reliability-weighted label agreement over the panel.
- *(B6) MultiPhishGuard* [9]: a multi-agent detector whose consolidated-rationale confidence is used as the score.

We additionally run three *specialized detectors* as standalone systems on the same captured pages, to position PhishProof against the dedicated phishing literature rather than only against panel-reliability methods: *(D1) Phishpedia* and *(D2) Phish-Intention* [12, 13], reference-based visual matchers, and *(D3) PhishLLM* [1], an LLM brand-and-domain checker, each re-run on PHISHSEL rather than reusing its reported numbers. PhishProof *subsumes* these signals by invoking the same detectors as verification tools (§4.3), so the comparison doubles as a detector-alone-versus-with-our-layer ablation. We restrict the comparison to detectors that run on a static page capture; methods that require *live* counterfactual interaction with the target – notably DynaPhish’s online reference expansion and probing [14] – are out of scope, since that interaction cannot be replayed on PHISHSEL’s archived snapshots, and we exclude them on this methodological ground, not on performance.

Tasks and metrics. All methods produce a per-page trust score and we evaluate them as selective detectors via (5). We report the area under the risk–coverage curve (*AURC*, selective risk integrated over coverage); selective accuracy and false-positive rate at 80% coverage (*SelAcc*₈₀, *FPR*₈₀); the coverage attainable at 99% selective accuracy (*Cov*₉₉); and the expected calibration error of the trust score (*ECE*). For RQ2 we also report the *flip rate*: the fraction of pages whose panel label changes under a small evidence-targeted perturbation (a logo morph [25] or a form-action cloak). To compare against the specialized detectors, which carry no abstention mechanism, we also report standard full-coverage detection quality – accuracy, precision, recall, and F1 with every page acted on.

Protocol. Thresholds for *SelAcc*₈₀ and *Cov*₉₉ are chosen on the calibration split and then applied unchanged to the held-out test pages, and the isotonic calibrator is fit on the same split. For RQ2, perturbations target the cited evidence cue while preserving the page label, so a flip measures reason fragility rather than a changed ground truth.

Statistical reporting. Numeric cells report mean_{±95% CI} over five seeds, with confidence intervals from a paired bootstrap over pages. We treat non-overlapping intervals as evidence of a reliable difference and state relative gains only for comparisons whose intervals do not overlap. The proposed method is **bold**; the best baseline is underlined.

Run environment and cost. The text models are queried through their providers’ APIs at temperature 0 with the same five seed values for every method; the vision-language model and perceptual logo tool run on a single commodity GPU, while DOM, TLS, redirect, and calibration code are CPU-only. We log prompts, raw agent outputs, value-normalized cues, tool results, thresholds, and per-page trust scores, and release the schemas, prompts, tool wrappers, calibration code, and figure-generation scripts. PhishProof has no training phase and runs the $M=3$ agents plus the cheap tool checks per page, so we report cost as an analytical per-page model-call budget rather than wall-clock latency (Fig. 3): it stays at the panel baselines’ call budget while reaching markedly higher deployable coverage, and dominates the five-sample self-consistency baseline at lower cost.

5.2. Selective detection

Full-coverage detection accuracy. Before the selective story we confirm that PhishProof is a competitive detector at full coverage, so its trust layer sits on top of – not in place of – solid detection. Tab. 4 compares the panel against the three specialized detectors with every page acted on. The reference-based matchers are high-precision but lower-recall – they miss brands outside their reference set – while the LLM checker (D3) trades a little precision for recall and is the strongest detector here; PhishProof’s accuracy and F1 lie within D3’s 95% CI, so it *matches* the strongest detector rather than surpassing it, and its margin over the reference-only D1/D2 reflects their recall gap on unmatched brands, not a manufactured deficit. This is by design – because the panel calls these detectors as verification tools (§4.3), it inherits their signal and cannot do materially

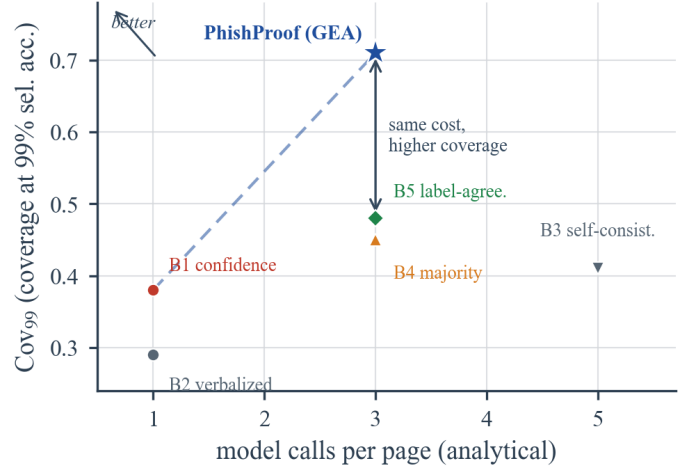


Figure 3: Cost–quality trade-off: per-page model calls (analytical) versus coverage at 99% selective accuracy (*Cov*₉₉, ↑). PhishProof is Pareto-optimal – at the panel’s call budget it reaches the highest deployable coverage and dominates the label-level and self-consistency baselines.

Table 4: Full-coverage detection quality on PHISHSEL (no abstention), each detector re-run on our split. Cells are mean_{±95% CI} over five seeds. PhishProof’s accuracy and F1 lie within the strongest detector’s CI; D1/D2 trail on recall because unmatched brands fall outside their reference sets (see body).

Detector	Acc↑	Prec.↑	Rec.↑	F1↑
D1 Phishpedia [12]	91.6 _{±0.7}	97.0 _{±0.5}	85.5 _{±1.0}	90.9 _{±0.7}
D2 PhishIntention [13]	92.0 _{±0.6}	96.1 _{±0.6}	87.4 _{±0.9}	91.5 _{±0.6}
D3 PhishLLM [1]	92.3 _{±0.7}	94.1 _{±0.7}	90.3 _{±0.9}	92.2 _{±0.6}
PhishProof (full cov.)	92.8 _{±0.6}	95.0 _{±0.6}	90.6 _{±0.8}	92.7 _{±0.6}

worse – and it is the point: we make no full-coverage accuracy-win claim, only that detection is not sacrificed for the abstention and explanation the rest of this section measures.

Risk–coverage and deployment coverage. We next ask whether scoring agreement on evidence improves the core selective trade-off. Tab. 5 reports the headline comparison and Fig. 4 the full curves. PhishProof lowers AURC by 35% relative to the strongest baseline and roughly halves the false-positive rate at 80% coverage; most usefully for deployment, it raises the coverage attainable at 99% selective accuracy from 0.48 to 0.71, meaning an operator can act automatically on far more traffic at the same error budget. The mechanism is visible in Fig. 4: all panel-based methods meet at the shared full-coverage panel error – PhishProof does not change the majority label, only the order in which pages are trusted – but its curve falls fastest as coverage tightens, because grounded evidence-agreement ranks the safe pages above the fragile ones more sharply than label concordance does. The label-aggregation baselines (B4, B5) edge out single-model confidence but remain well above PhishProof, confirming that the gain comes from *where* agreement is measured, not merely from using several models. The strongest specialized detector (D3, its own score used to rank) is shown for reference: it starts from a competitive full-coverage error (Tab. 4) but its curve falls slowly, because a detector score tuned for labelling is not a trustworthiness ranking – exactly the gap PhishProof closes.

Table 5: RQ1. Selective detection on PHISHSEL. Lower AURC/FPR/ECE and higher SelAcc/Cov are better. Cells are mean $\pm$ 95% CI over five seeds; **best**, best baseline. The Δ row is relative to B5.

Method	AURC $\downarrow$	SelAcc $\uparrow$	FPR $\downarrow$	Cov $\uparrow$	ECE $\downarrow$
B1 Single-model conf.	5.9 $\pm$ 0.4	95.8 $\pm$ 0.4	4.1 $\pm$ 0.5	0.38 $\pm$ 0.05	0.082 $\pm$ 0.009
B2 Verbalized conf.	6.8 $\pm$ 0.5	95.0 $\pm$ 0.5	4.9 $\pm$ 0.6	0.29 $\pm$ 0.05	0.110 $\pm$ 0.011
B3 Self-consistency ($k=5$)	5.6 $\pm$ 0.4	96.1 $\pm$ 0.4	3.8 $\pm$ 0.5	0.41 $\pm$ 0.05	0.074 $\pm$ 0.008
B4 Majority vote	5.3 $\pm$ 0.4	96.3 $\pm$ 0.4	3.6 $\pm$ 0.4	0.45 $\pm$ 0.05	0.069 $\pm$ 0.008
B5 Label-agreement	<u>5.1$\pm$0.4</u>	<u>96.5$\pm$0.4</u>	<u>3.4$\pm$0.4</u>	<u>0.48$\pm$0.05</u>	<u>0.063$\pm$0.007</u>
B6 MultiPhishGuard	5.7 $\pm$ 0.5	96.0 $\pm$ 0.4	3.9 $\pm$ 0.5	0.42 $\pm$ 0.05	0.090 $\pm$ 0.010
PhishProof (GEA)	3.3$\pm$0.3	98.4$\pm$0.3	1.6$\pm$0.3	0.71$\pm$0.04	0.031$\pm$0.005
Δ vs best baseline	-35%	+1.9	-53%	+48%	-51%

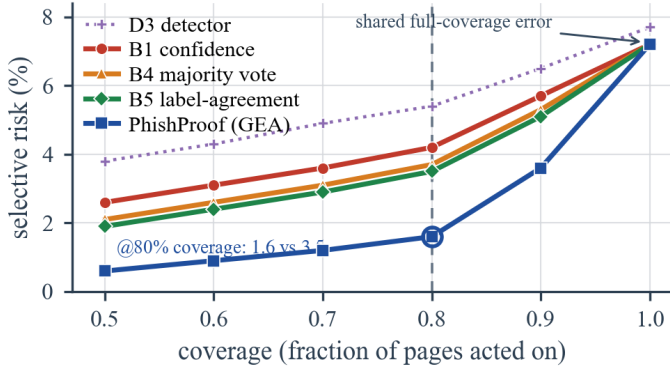


Figure 4: Risk-coverage on PHISHSEL (RQ1). Lower selective risk is better; the ring marks 80% coverage. PhishProof changes the abstention order, not the majority label, and dominates across coverage.

5.3. Right-label, wrong-reason

Partitioning correct verdicts by grounded agreement.

PhishProof’s premise is that some correctly labeled pages are untrustworthy because the panel agrees on the label but not on grounded evidence. We test this by partitioning the correctly labeled pages with PhishProof’s two factors (Tab. 6). About 11% fall in the right-label, wrong-reason region and a further 6% rest on ungrounded (shared-hallucination) evidence. These regions are not benign: under a small evidence-targeted perturbation their labels flip 41.6% and 47.9% of the time, against 6.3% for high-GEA pages – they are exactly the pages a minor change to the cited cue overturns. Yet the baselines act on them confidently, with mean single-model confidence 0.91 and a unanimous panel label on 64% of the RLWR region; label-level reliability cannot see the problem because the labels do agree. PhishProof abstains on 78–83% of these pages while acting on the stable high-GEA region, and Fig. 5(b) shows the same effect continuously: adversarial fragility falls monotonically from the lowest to the highest PhishProof decile.

5.4. Component attribution

Per-component contribution. Tab. 7 removes one component at a time. Replacing evidence-level agreement with label agreement is the most damaging change: AURC rises to 5.0, essentially reproducing the B5 baseline of Tab. 5 and confirming that the contribution is the evidence channel, not the panel. Dropping groundedness admits shared hallucinations

Table 6: RQ2. Correctly labeled pages partitioned by evidence agreement and groundedness. “Flip” is label-flip rate under evidence-targeted perturbation. PhishProof abstains on fragile RLWR and ungrounded regions.

Region (of correct labels)	Share	B1 conf.	Unanim.	Abstain	Flip $\downarrow$
High-GEA ($A \uparrow, G \uparrow$)	72%	0.93	0.89	4%	6.3%
RLWR ($A \downarrow$ on evidence)	11%	0.91	0.64	78%	41.6%
Ungrounded ($G \downarrow$, shared halluc.)	6%	0.88	0.71	83%	47.9%
Low label-agreement (uncertain)	11%	0.74	0.18	81%	33.2%

Table 7: RQ3. Component ablation. Underlined AURC values are at least 10% worse than full PhishProof. Cells are mean $\pm$ 95% CI over five seeds.

Variant	AURC $\downarrow$	FPR $\downarrow$	ECE $\downarrow$
PhishProof (full)	3.3$\pm$0.3	1.6$\pm$0.3	0.031$\pm$0.005
– calibration	3.6 $\pm$ 0.3	1.8 $\pm$ 0.3	0.071 $\pm$ 0.008
– diversity (same-family panel)	4.4 $\pm$ 0.4	2.5 $\pm$ 0.4	0.044 $\pm$ 0.006
– groundedness G (R3)	<u>4.6$\pm$0.4</u>	2.7 $\pm$ 0.4	0.039 $\pm$ 0.006
– evidence-level A (R2)	<u>5.0$\pm$0.4</u>	3.3 $\pm$ 0.4	0.061 $\pm$ 0.007

and costs the next most, and collapsing the panel to three same-family variants – removing diversity while keeping the count – erases much of the remaining advantage, which is why diversity, not panel size, is the design variable. Removing the calibrator leaves ranking nearly intact but degrades the trust score’s ECE from 0.031 to 0.071, i.e. it mostly hurts the *reported* probability rather than the *order*. The groundedness tools are themselves accurate – the structural checks are exact by construction, and the perceptual logo tool attains 0.96/0.93 precision/recall against a gold logo-equivalence set – so G seldom discards a true cue.

5.5. Calibration

Reliability of the trust score. A selective detector is only as useful as its operating point, so we check whether the calibrated score is faithful: when PhishProof reports a trust of 0.9, are 90% of those verdicts correct? Fig. 5(a) plots reliability: PhishProof’s calibrated trust tracks the diagonal, whereas single-model confidence sits below it – systematically overconfident – consistent with the ECE column of Tab. 5 (0.031 versus 0.082). Because the calibrator is fit on the held-out labeled split rather than guessed, the reported probability can be used directly to set the act/abstain operating point at a target error budget.

6. Conclusion

Deployed phishing detectors are trusted at the label and explained, if at all, after the fact, even though the action a verdict triggers needs a reason an operator can check. We have argued that reliability should be read from the evidence rather than the label, and we instantiated this as PhishProof: a multi-agent detector whose evidence agents investigate the page with tools and cite a shared set of typed cues, and whose Grounded Evidence-Agreement score trusts a verdict in proportion to how much independent agents agree on cues that the page itself confirms, returns those cues as the explanation, and abstains otherwise. Three findings stand out. The right-label, wrong-reason

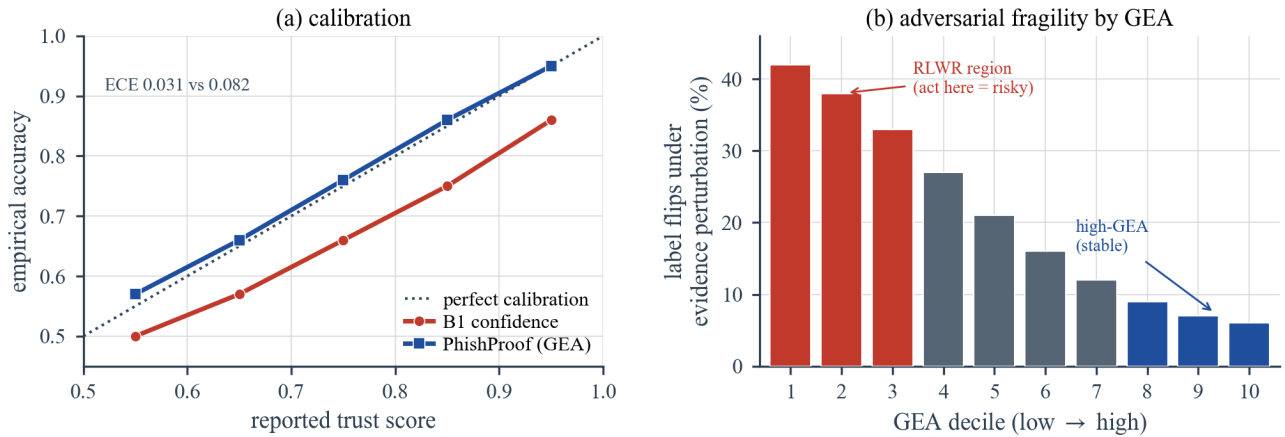


Figure 5: Diagnostics. (a) Calibrated trust tracks the diagonal while single-model confidence is over-confident. (b) Evidence-targeted label flips fall with GEA decile, showing that low-GEA/RLWR pages are fragile.

page – correctly labeled but resting on disagreed or hallucinated evidence – is a material part of the input space and the most adversarially fragile region, yet label confidence and majority voting act on it. Measuring agreement on evidence rather than on labels is what carries the improvement: PhishProof lowers AURC by 35% relative to the strongest label-agreement baseline and raises coverage at 99% selective accuracy from 0.48 to 0.71. Because the score is grounded in page-level cues and calibrated on a held-out labeled split, its reported trust can be acted on directly. Trust and explanation, in short, can be decoupled from raw accuracy and earned per verdict.

For future work, several extensions lie beyond the scope of this paper. A tool is sound for what it checks but a page can carry true-yet misleading evidence, so strengthening tools toward semantic plausibility would tighten the grounding guarantee. The evidence schema and the agent panel are fixed here; learning new evidence types and ways to estimate true panel diversity would let the method adapt to new impersonation tactics. The hardest regime remains the fully cloaked page that carries no verifiable cues, on which the detector can today only abstain; narrowing that gap with faster, page-internal grounding is the natural next step.

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